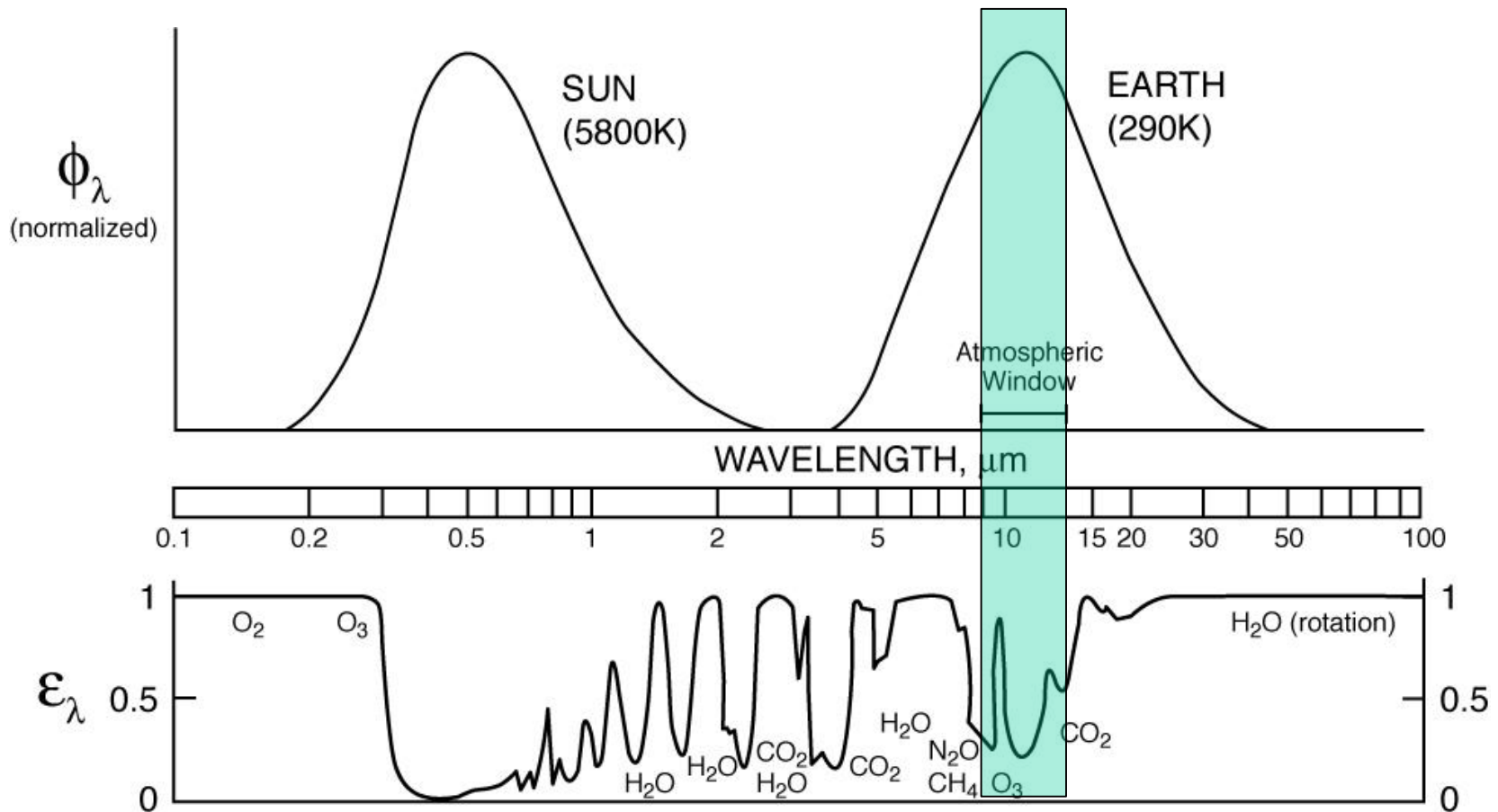
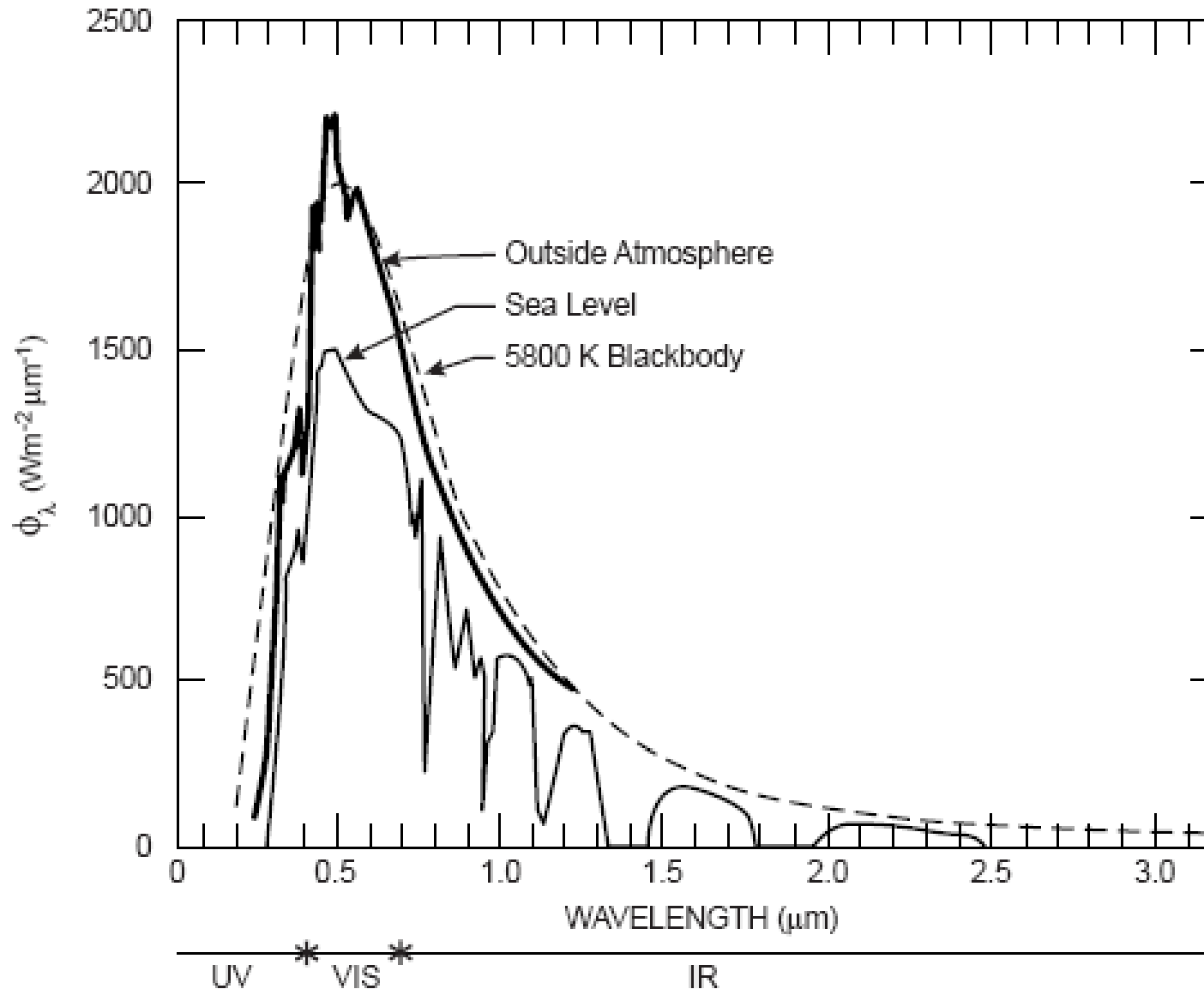


GREENHOUSE EFFECT: absorption of terrestrial radiation by the atmosphere



- Major greenhouse gases: H_2O , CO_2 , CH_4 , O_3 , N_2O , CFCs,...
- Not greenhouse gases: N_2 , O_2 , Ar, ...

SOLAR RADIATION SPECTRA: IN SPACE AND AT SEA LEVEL



**SOLAR
SPECTRA
APPROX.
EQUIV
TO THAT
OF BLACK
BODY
RADIATION
AT 5777 K**

KIRCHOFF'S LAW

- The Planck black body formulation for emission of radiation can be generalized to all objects using Kirchhoff's law

If an object absorbs radiation of wavelength λ with an efficiency ε_λ , then it emits radiation of that wavelength at a fraction ε_λ of the corresponding blackbody emission at the same temperature.

Using Kirchhoff's law & Planck's relation, one can derive the emission spectrum of any object by simply knowing its absorption spectrum & its temperature

$$\phi_\lambda(T) = \varepsilon_\lambda(T) \phi_\lambda^b(T) \quad \text{KIRCHOFF'S LAW}$$

ABSORPTION OF RADIATION BY GAS MOLECULES

- ...requires quantum transition in internal energy of molecule.
- THREE TYPES OF TRANSITION
 - **Electronic transition: UV radiation ($<0.4\ \mu\text{m}$)**
 - Jump of electron from valence shell to higher-energy shell, sometimes results in dissociation (example: $\text{O}_3 + h\nu \rightarrow \text{O}_2 + \text{O}$)
 - **Vibrational transition: near-IR ($0.7\text{-}20\ \mu\text{m}$)**
 - Increase in vibrational frequency of a given bond requires change in dipole moment of molecule
 - **Rotational transition: far-IR ($20\text{-}100\ \mu\text{m}$)**
 - Increase in angular momentum around rotation axis

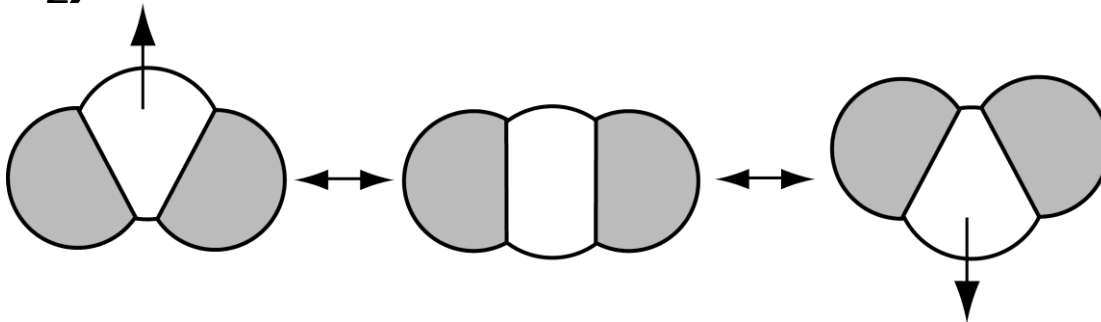
Gases that absorb radiation near the spectral maximum of terrestrial emission ($10\ \mu\text{m}$) are called *greenhouse gases*; this requires vibrational or vibrational-rotational transitions

Infrared spectroscopy (1)

$$E_{total} = E_{vib} + E_{rot} + E_{trans}$$

vibrational, rotational, translational transitions

- **Permanent or temporary oscillating dipole (e.g. CO₂)**



BLACK BODY RADIATION

- The monochromatic emissive power of a black body “ ϕ_λ ” is given by the **Planck function**:

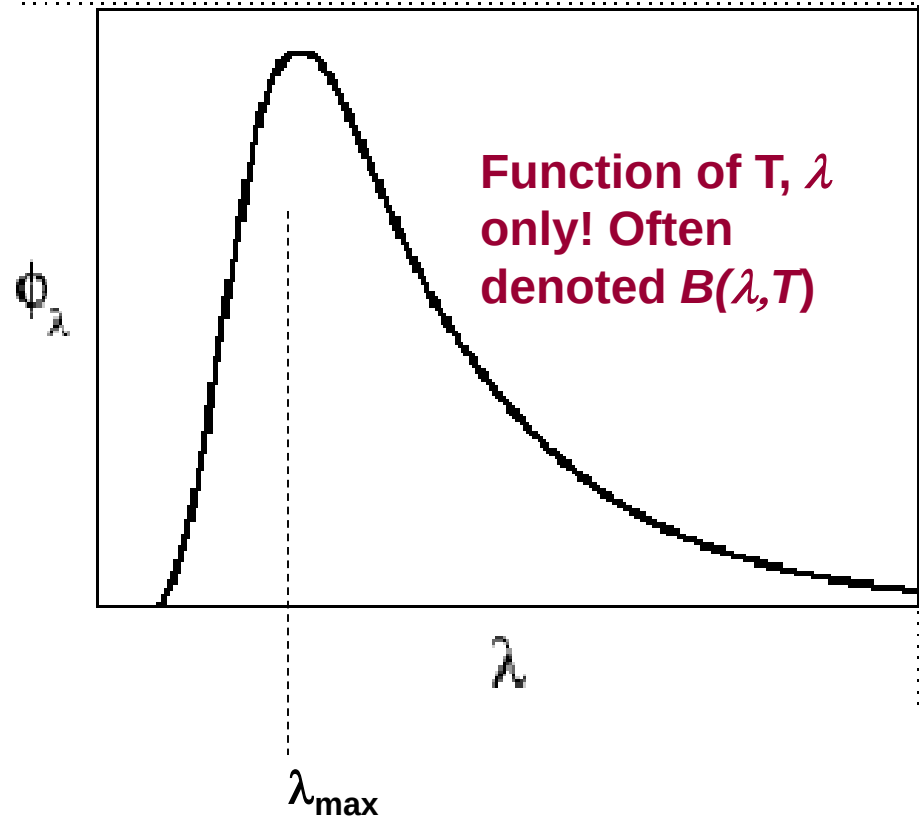
$$\phi_\lambda^b = \frac{2\pi hc^2}{\lambda^5 \left(\exp\left(\frac{hc}{kT\lambda}\right) - 1 \right)}$$

When integrated over all λ ,

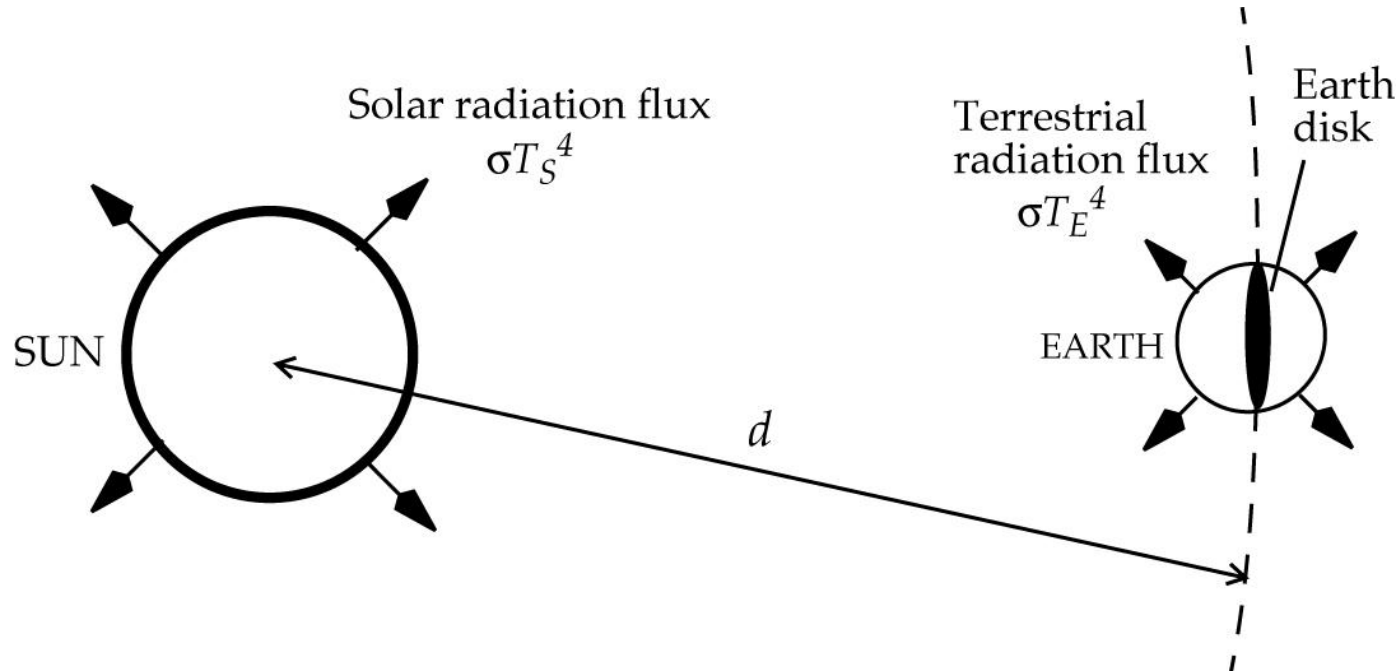
$$\Phi = \sigma T^4$$

$\sigma = 2\pi^5 k^4 / 15c^2 h^3$ is the **Stefan-Boltzmann constant**

$$\lambda_{\max} = hc/5kT \quad \text{Wien's law}$$



RADIATIVE EQUILIBRIUM FOR THE EARTH



Solar radiation flux intercepted by Earth = **solar constant $F_S = 1370 \text{ W m}^{-2}$**

Radiative balance $\Rightarrow$ **effective temperature** of the Earth:

$$T_E = \left[\frac{F_S(1 - A)}{4\sigma} \right]^{\frac{1}{4}} = 255 \text{ K}$$

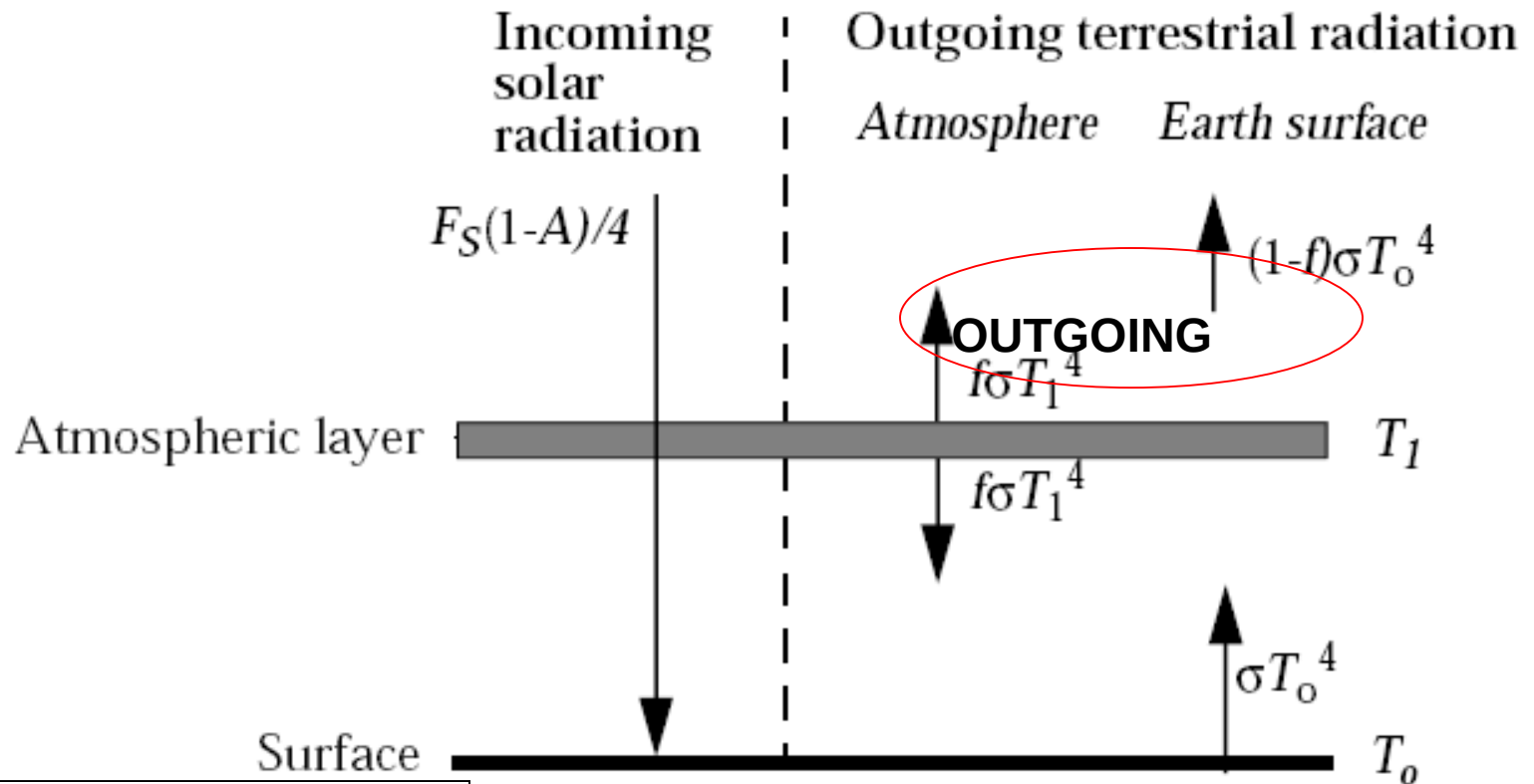
where A is the **albedo** (reflectivity) of the Earth

Planetary radiation balance

Planet	Distance to Sun (AU)*	S_0 (W m ⁻²)	A	T_e (K)	T_s (K)
Mercury	0.39	9007	0.08	437	437
Venus	0.72	2640	0.75	232	732
Earth	1.0	1366	0.3	255	288
Mars	1.52	590	0.15	217	223

*** Astronomical Units: 1 AU $\approx$ 1.5 $\times$ 10⁸ km**

SIMPLE MODEL OF GREENHOUSE EFFECT



Energy balance equations:

- *Earth system*

$$F_S(1-A)/4 = (1-f)\sigma T_o^4 + f\sigma T_1^4$$

- *Atmospheric layer*

$$f\sigma T_o^4 = 2f\sigma T_1^4$$

Solution:

$$T_o = \left[\frac{F_S(1-A)}{4(1-\frac{f}{2})\sigma} \right]^{\frac{1}{4}} \begin{matrix} T_o = 288 \text{ K} \\ \Rightarrow f = 0.77 \\ T_1 = 241 \text{ K} \end{matrix}$$

Question

Why do N_2 and O_2 not contribute to the greenhouse effect?

Answer: no dipole moment

WHAT IS RADIATIVE FORCING ?

- GCMs are used to estimate the surface warming associated with an increase in GHGs
- In these models, radiative perturbation (e.g. by adding a GHG triggers an initial warming; complex responses follow (e.g. more evapo-transpiration, circulation change)
- Radiative forcing due to a GHG X is defined as the flux imbalance in the radiative budget of the Earth system caused by adding a mass Δm of X to the atmospheric mass of the GHG X

Numerically,

$$\Delta F = F_{\text{in}} - F_{\text{out}}$$

RADIATIVE FORCING & SURFACE TEMP.

As shown in the prev. Figure, suppose adding Δm of GHG corresponds to increase of Δf in absorption efficiency of atmosphere, then :

$$\Delta F = \left(1 - \frac{f}{2}\right) \sigma T_o^4 - \left(1 - \frac{f + \Delta f}{2}\right) \sigma T_o^4 = \frac{\Delta f}{2} \sigma T_o^4$$

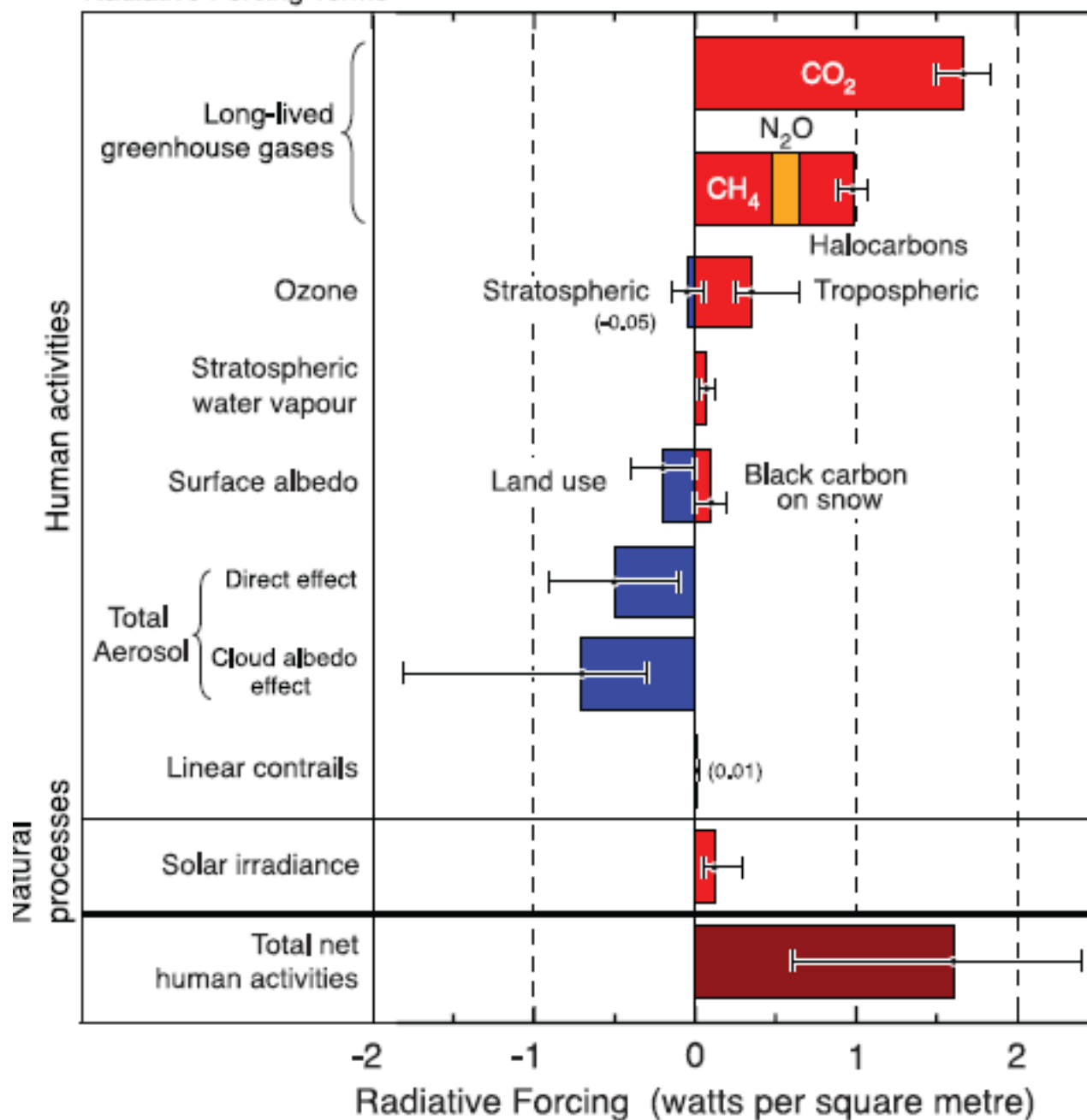
**Radiative
forcing**

**Outgoing
Terrestrial
radiation for initial
atmosphere in radiative
eqb.**

**Outgoing
Terrestrial
radiation for perturbed
atmosphere**

Radiative forcing of climate between 1750 and 2005

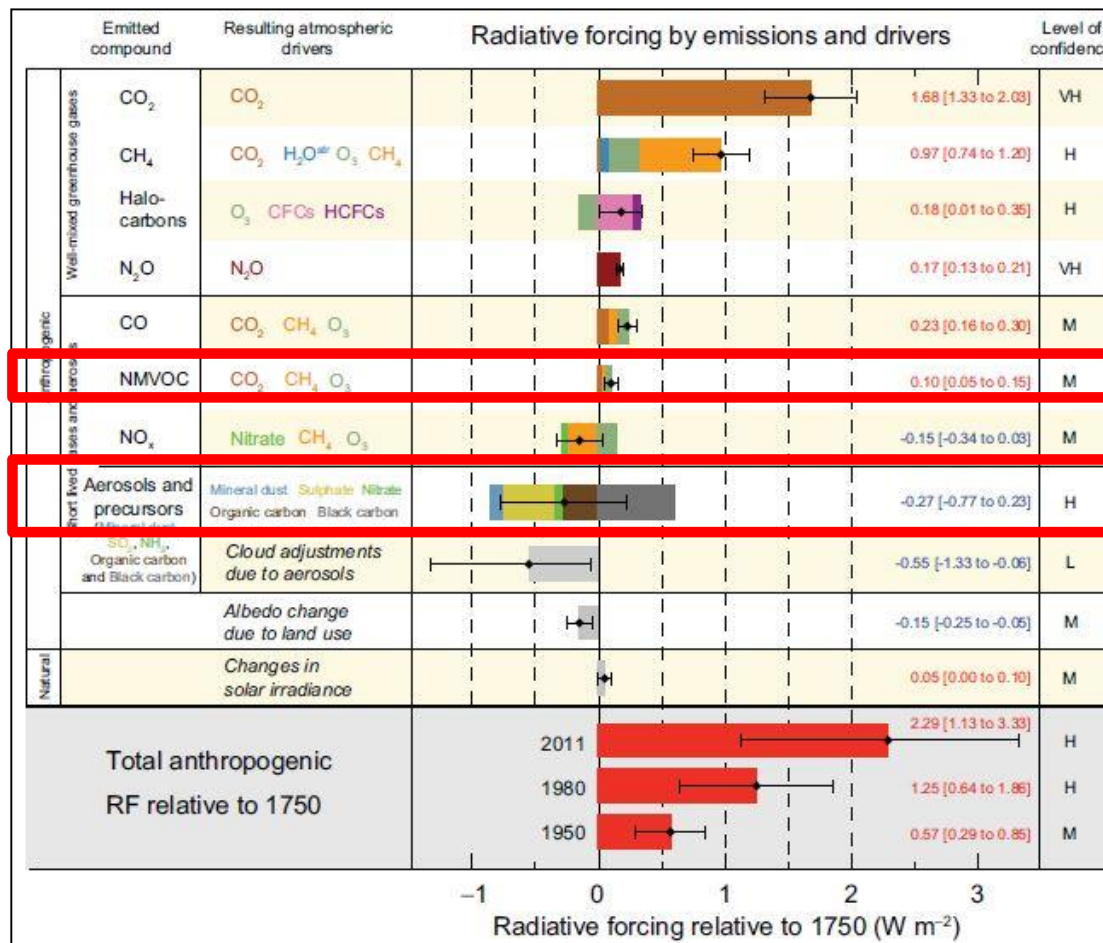
Radiative Forcing Terms



IPCC [2007]

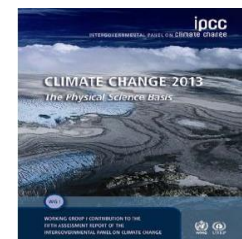
VOCs: Relevance to Climate

- VOC definition as per IUPAC: “carbon containing compounds which possess high vapour pressure (> 10 Pa at 25°C) and low boiling point ($< 260^{\circ}\text{C}$) at standard atmospheric pressure (101.3 kPa)”.
- It includes alcohols, aldehydes, esters, acids, nitriles, amines, amides, halocarbons and organosulphur compounds, excluding carbon monoxide (CO) and carbon dioxide (CO_2)



- Level of scientific understanding for VOCs is medium and for aerosol is high.
- Therefore scientific studies related to VOCs become more important and needs to be carried out.

AR5,
IPCC [2013]



RADIATIVE FORCING AND G.W.P.

- By computing radiative forcings associated with changes in emissions of individual GHG, we can assess and compare the potential climate effects of different gases & make appropriate policy decisions
- International environmental policy on GHGs emissions makes use of the Global Warming Potential (G.W.P.) index, which is defined as the radiative forcing resulting from an instantaneous 1 kg injection of a given gas into the atmosphere relative to the radiative forcing from an Instantaneous 1 kg injection of CO₂:

$$GWP = \frac{\int_{t_0}^{t_0 + \Delta t} \Delta F_{1kg \text{ X}} dt}{\int_{t_0}^{t_0 + \Delta t} \Delta F_{1kg \text{ CO}_2} dt}$$

Table 7-1 Global warming potentials from the instantaneous injection of 1 kg of a trace gas, relative to carbon dioxide

Gas	Lifetime, years	Global warming potential over integration time horizon		
		20 years	100 years	500 years
CO ₂	~100	1	1	1
CH ₄	10	62	25	8
N ₂ O	120	290	320	180
CFC-12	102	7900	8500	4200
HCFC-123	1.4	300	93	29
SF ₆	3200	16500	24900	36500

Table 7-1 lists GWPs for several greenhouse gases and different time horizons. The synthetic gases CFCs, hydrofluorocarbons (HFCs) such as HCFC-123, and SF₆ have large GWPs because they absorb in the atmospheric window. The GWP of HFCs is less than that of CFCs because HFCs have shorter atmospheric lifetimes.

Question

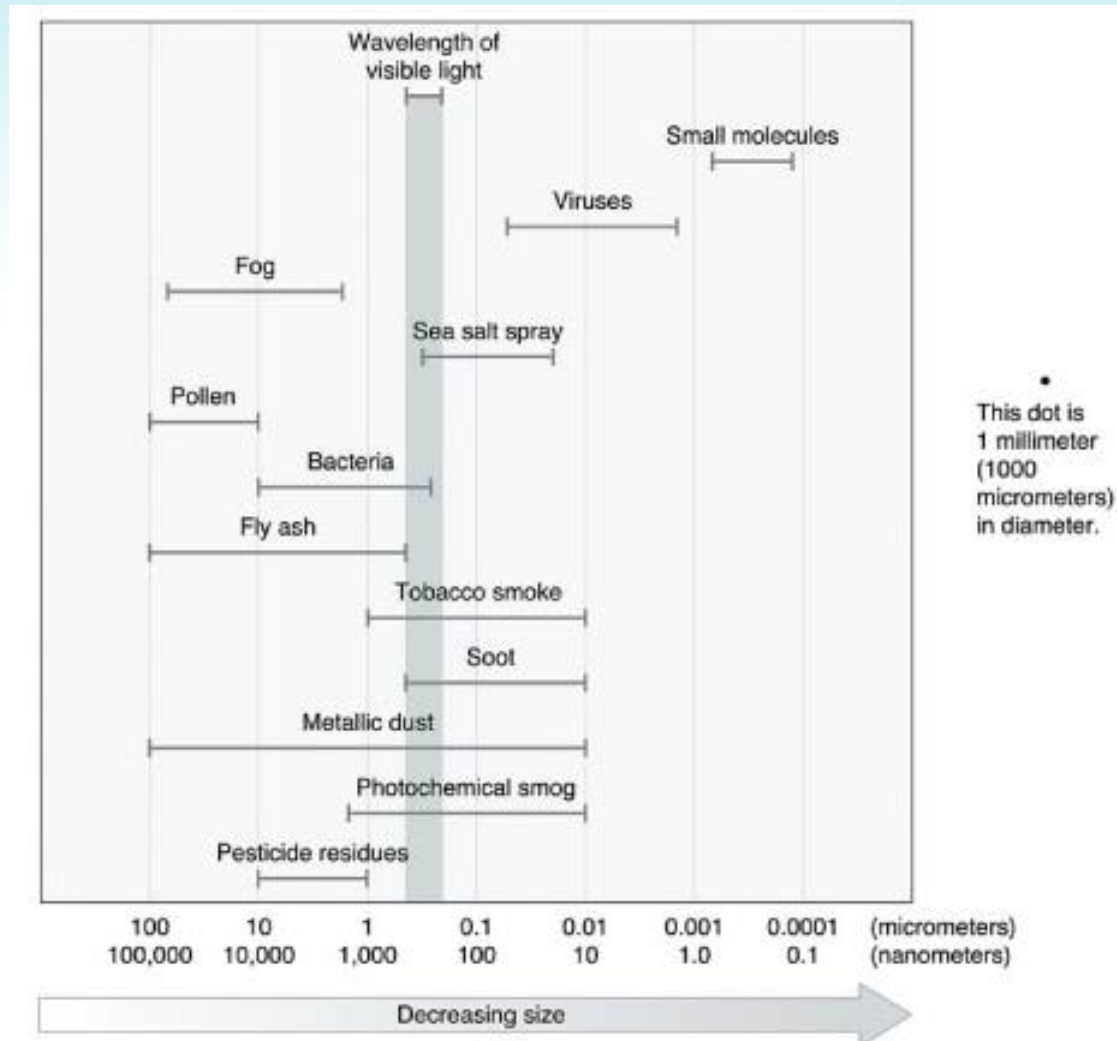
At what λ does Earth radiate most energy?

At TOA (i.e. seen from space) $T_e = 255$ K

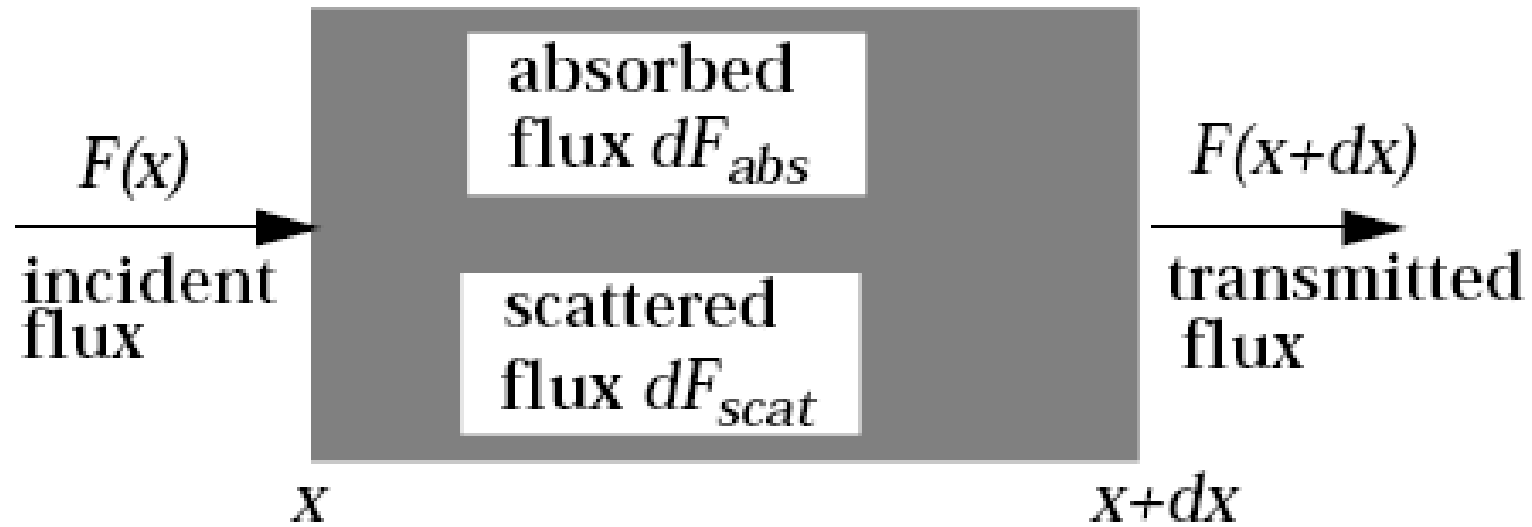
Answer:

$$\lambda_{\max} = \frac{hc}{4.965kT} = \mathbf{11.4 \mu m}$$

Figure 2.7 Suspended particulate matter of many types and sizes enters the atmosphere from both natural and anthropogenic sources.



Optical Depth



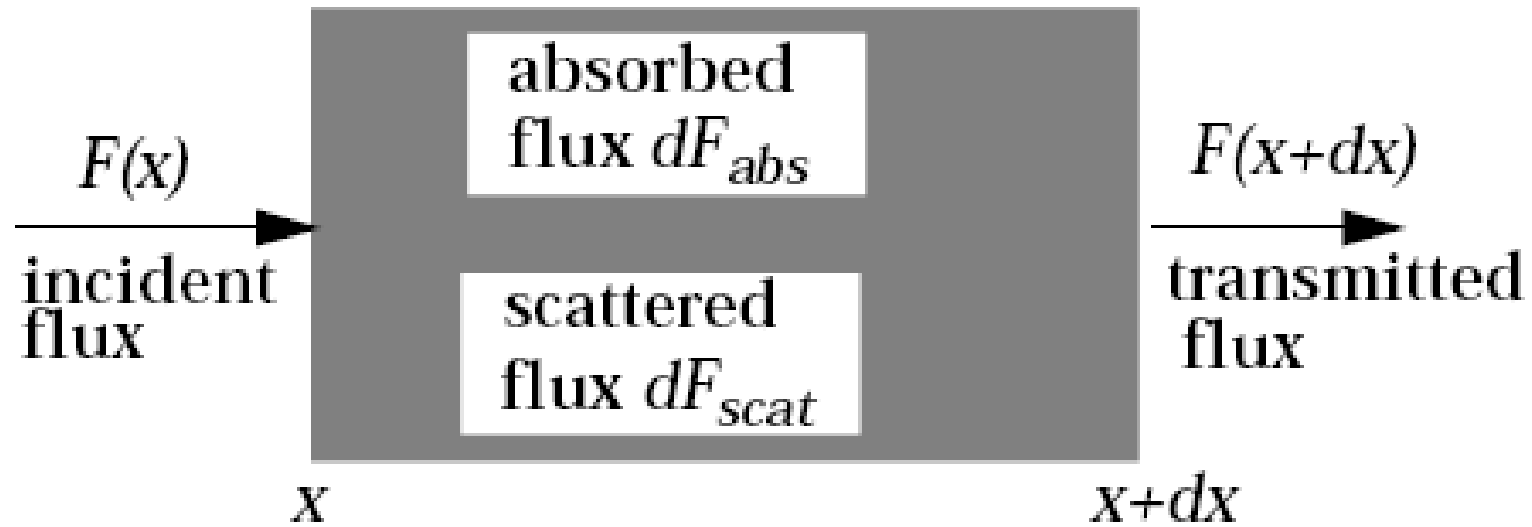
$$F(x + dx) = F(x) - dF_{abs} - dF_{scat}$$

We expect dF_{abs} & dF_{scat} to be proportional to $F(x)$, dx and the number density n of the absorber or scatterer in the slab. Thus,

$$dF_{abs} = n\sigma_{abs}F(x)dx$$

$$dF_{scat} = n\sigma_{scat}F(x)dx$$

Optical Depth



$$dF = F(x + dx) - F(x) = -n(\sigma_{abs} + \sigma_{scat})Fdx$$

Integrating we L.H.S. and R.H.S. we obtain,

$$F(L) = F(0) \exp[-n(\sigma_{abs} + \sigma_{scat})L]$$

**Note radiation intensity $F(x)$
decreases exponentially with L**

**$\delta = n(\sigma_{abs} + \sigma_{scat})L$ is defined as the optical depth
of the slab**

Total Optical Depth of Aerosols

If the slab contains k different absorbers or scatterers, The Total Optical Depth is obtained by adding the contributions of all species:

$$\delta_T = \sum_k \delta_i = \sum_k n_i (\sigma_{abs,i} + \sigma_{scat,i}) L$$

SCATTERING OF RADIATION BY AEROSOLS: “DIRECT EFFECT”

Scattering efficiency is maximum when particle diameter = λ
⇒ particles in 0.1-1 μm size range are efficient scatterers of solar radiation

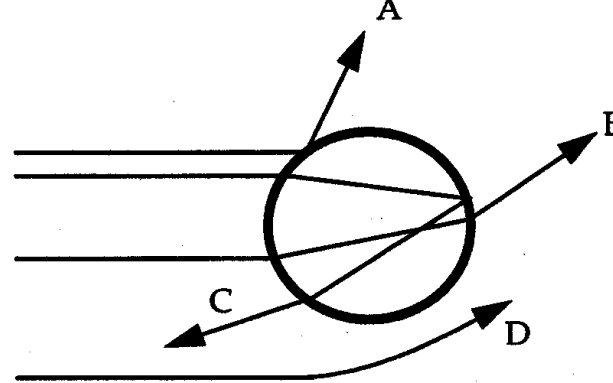


Fig. 8-3 Scattering of a radiation beam: processes of reflection (A), refraction (B), refraction and internal reflection (C), and diffraction (D).

By scattering solar radiation, aerosols increase the Earth's albedo

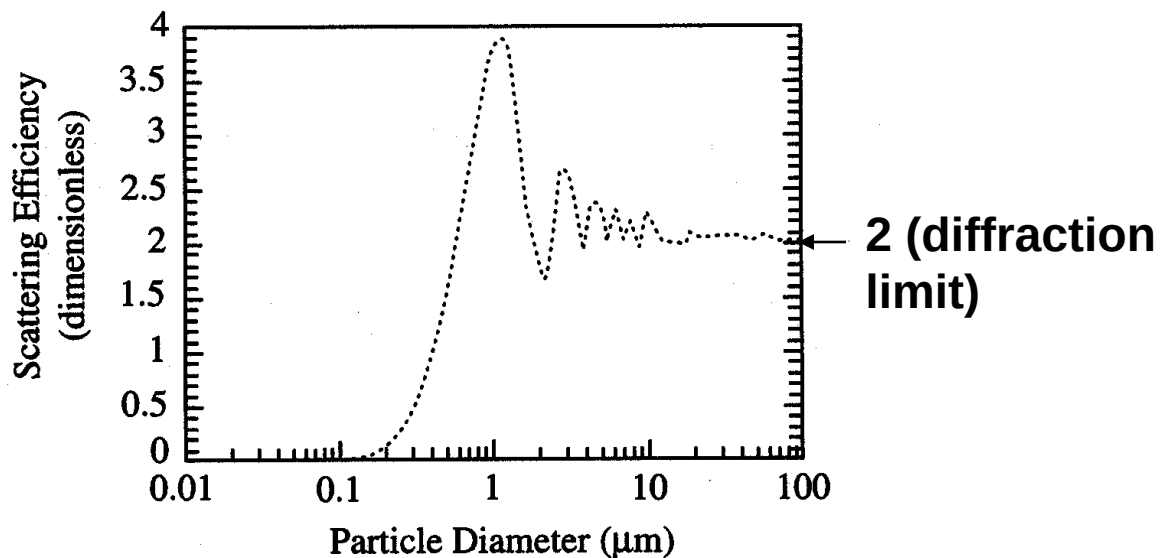


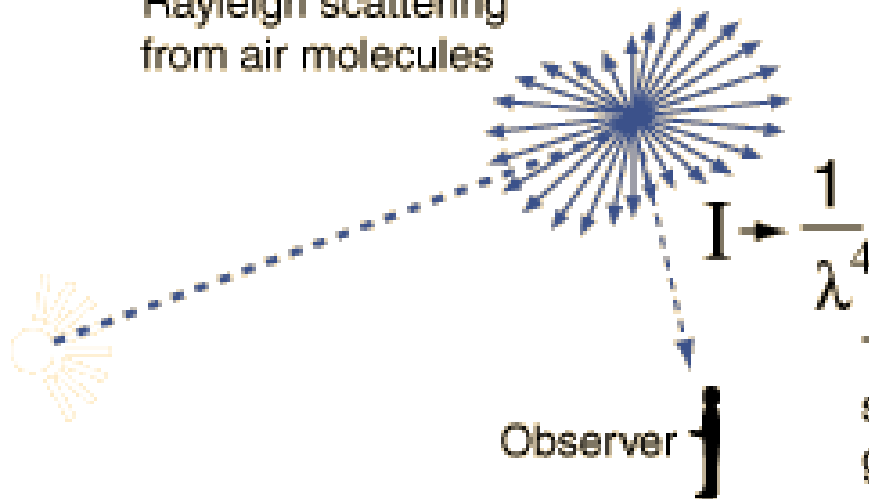
Fig. 8-4 Scattering efficiency of green light ($\lambda = 0.5 \mu\text{m}$) by a liquid water sphere as a function of the diameter of the sphere. Scattering efficiencies can be larger than unity because of diffraction. Adapted from Jacobson, M. Z. *Fundamentals of Atmospheric Modeling*. Cambridge, England: Cambridge University Press, 1999.

RAYLEIGH SCATTERING

$$I = I_0 \frac{8\pi^4 N\alpha^2}{\lambda^4 R^2} (1 + \cos^2 \theta)$$

Scattering at right angles
is half the forward intensity
for Rayleigh scattering

Rayleigh scattering
from air molecules



N = # of scatterers

α = polarizability

R = distance from scatterer

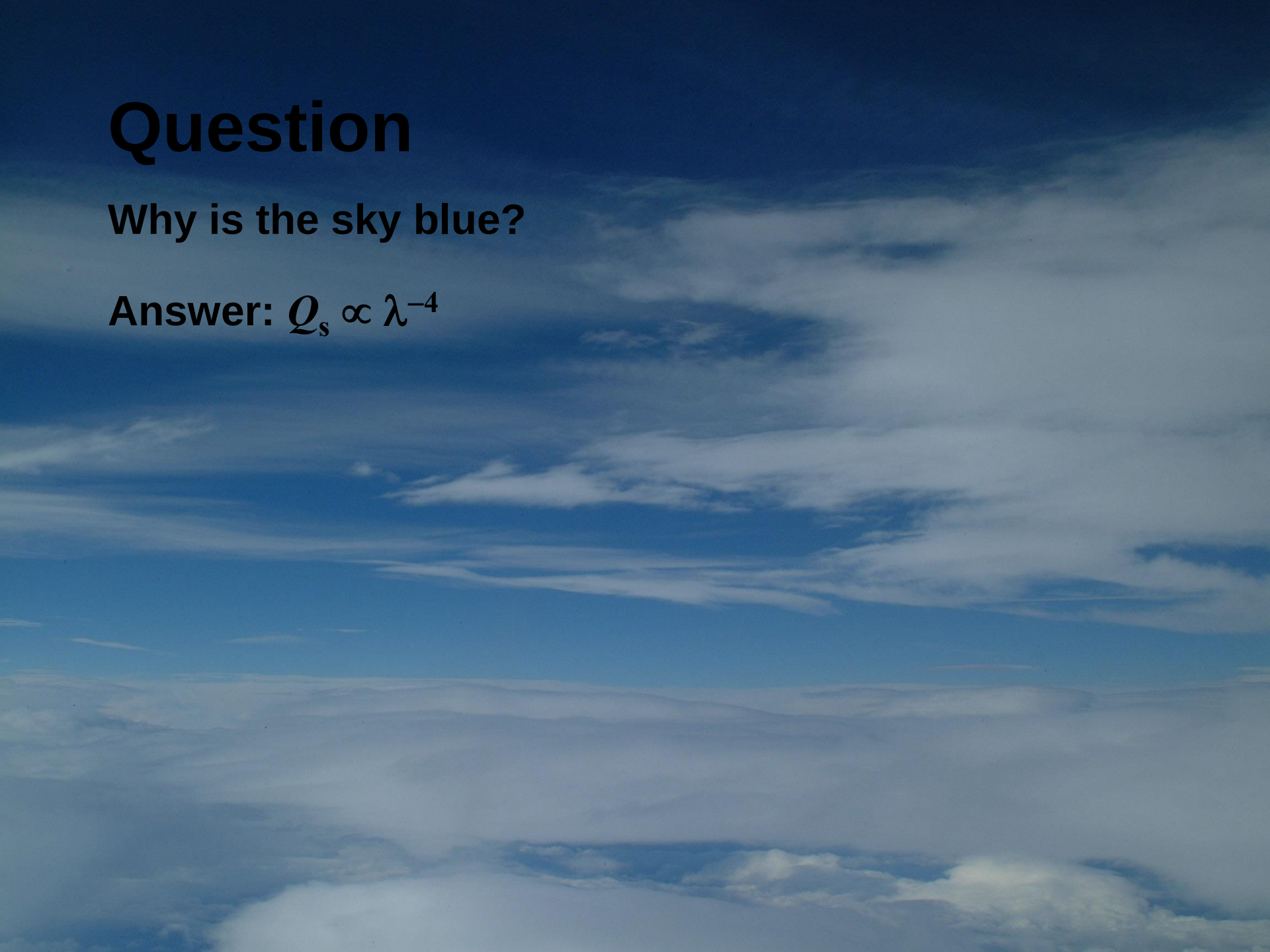
The strong wavelength dependence of Rayleigh scattering enhances the short wavelengths, giving us the blue sky.

Most effective for particles $< \lambda/10$

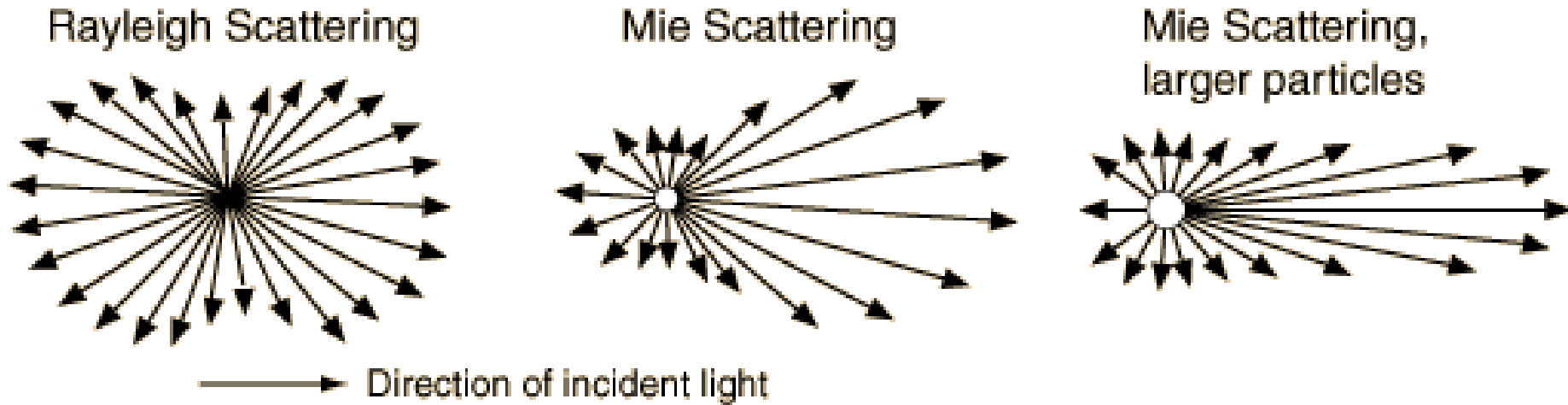
Question

Why is the sky blue?

Answer: $Q_s \propto \lambda^{-4}$



MIE SCATTERING



**Most effective for particles $\geq \lambda$
e.g. water droplets in clouds!**

Question

Why are clouds white?

Answer: $Q_s \propto \lambda^2$



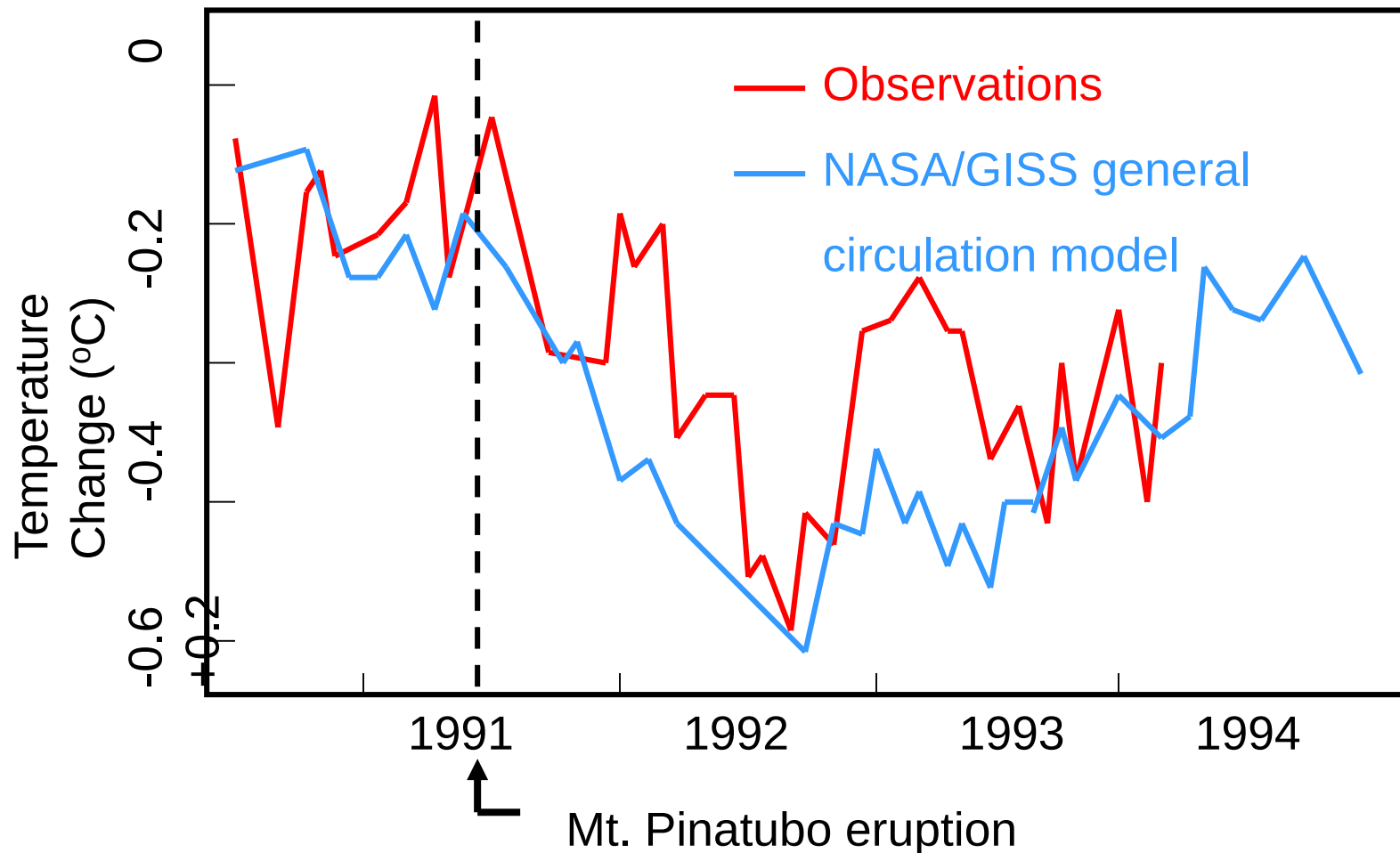
Question

Why are clouds bright?

Answer: forward scattering

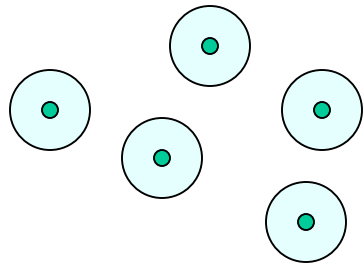
EVIDENCE OF AEROSOL EFFECTS ON CLIMATE:

Temperature decrease following large volcanic eruptions



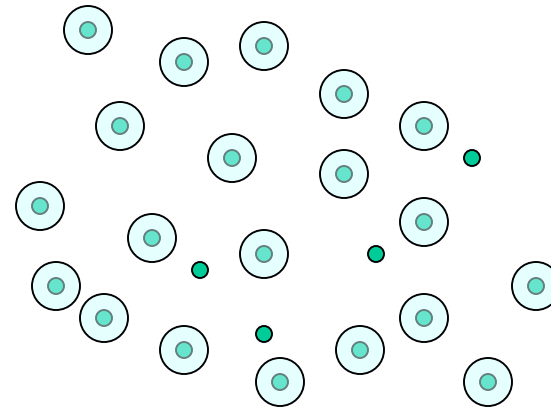
AEROSOL “INDIRECT EFFECT” FROM CLOUD CHANGES

Clouds form by condensation on preexisting aerosol particles (“cloud condensation nuclei”) when $RH > 100\%$



clean cloud (few particles):
large cloud droplets

- low albedo
- efficient precipitation



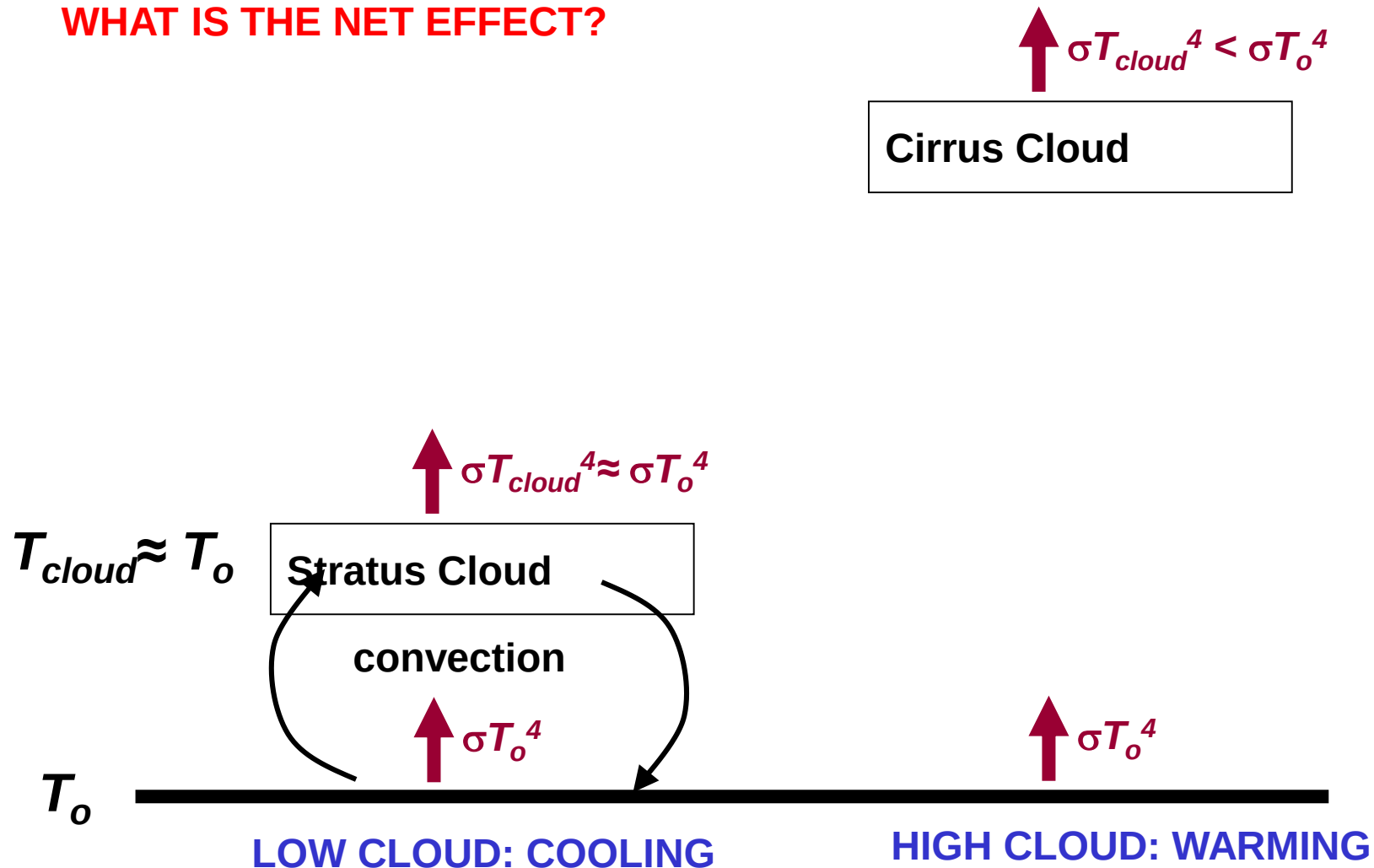
polluted cloud (many particles):
small cloud droplets

- high albedo
- suppressed precipitation

CLIMATE FEEDBACK FROM HIGH vs. LOW CLOUDS

Clouds reflect solar radiation ($\Delta A > 0$) → cooling;
...but also absorb IR radiation ($\Delta f > 0$) → warming

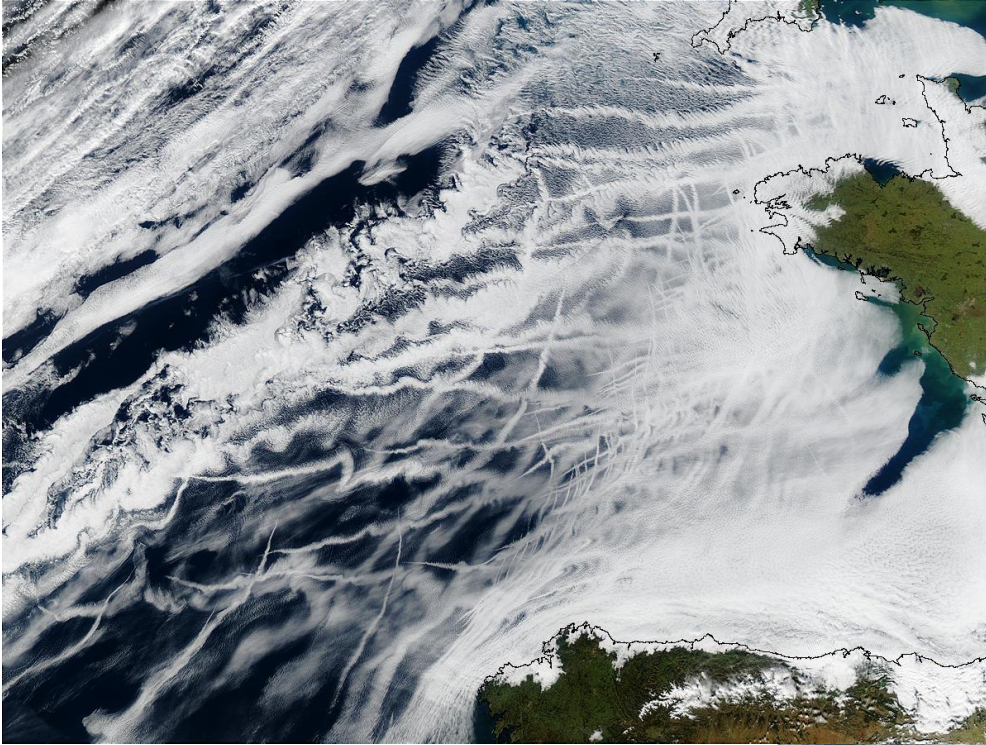
WHAT IS THE NET EFFECT?



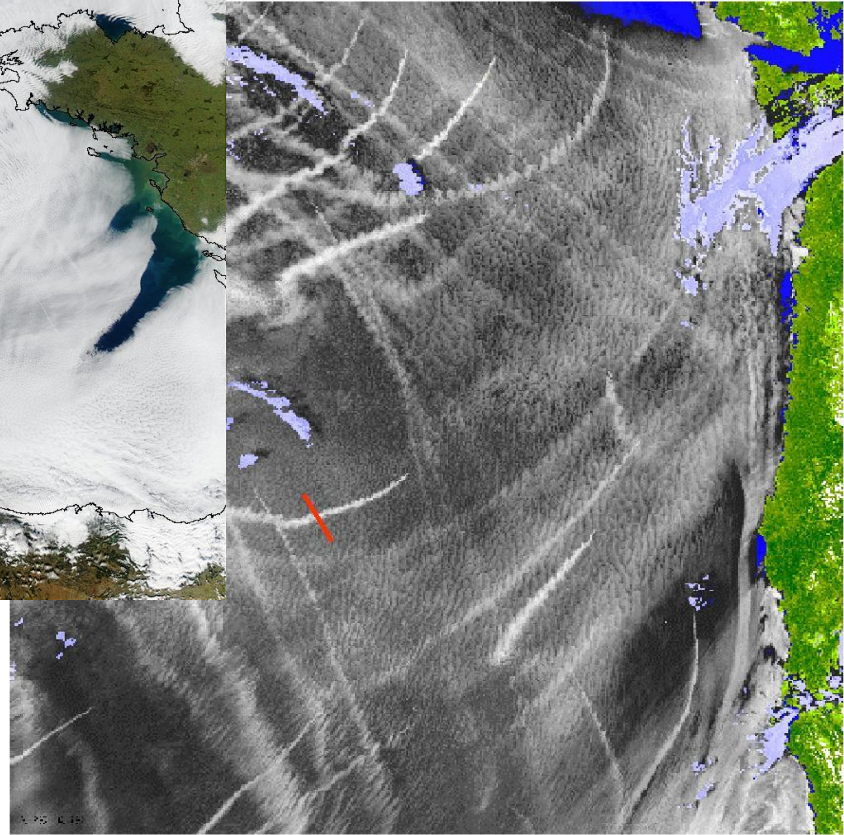
Cloud Cover: Impact on Climate

- **Feedbacks associated with changes in cloud cover represent the largest uncertainty in current estimates of climate change**
- **Clouds not only increase the albedo of the Earth, they are also efficient absorbers of IR radiation & hence contribute to the Green house effect**
- **High Clouds (e.g. Cirrus) cause net heating while low clouds cause net cooling**
- **Ship tracks and aircraft contrails perturb the radiation budget as well**

SATELLITE IMAGES OF SHIP TRACKS

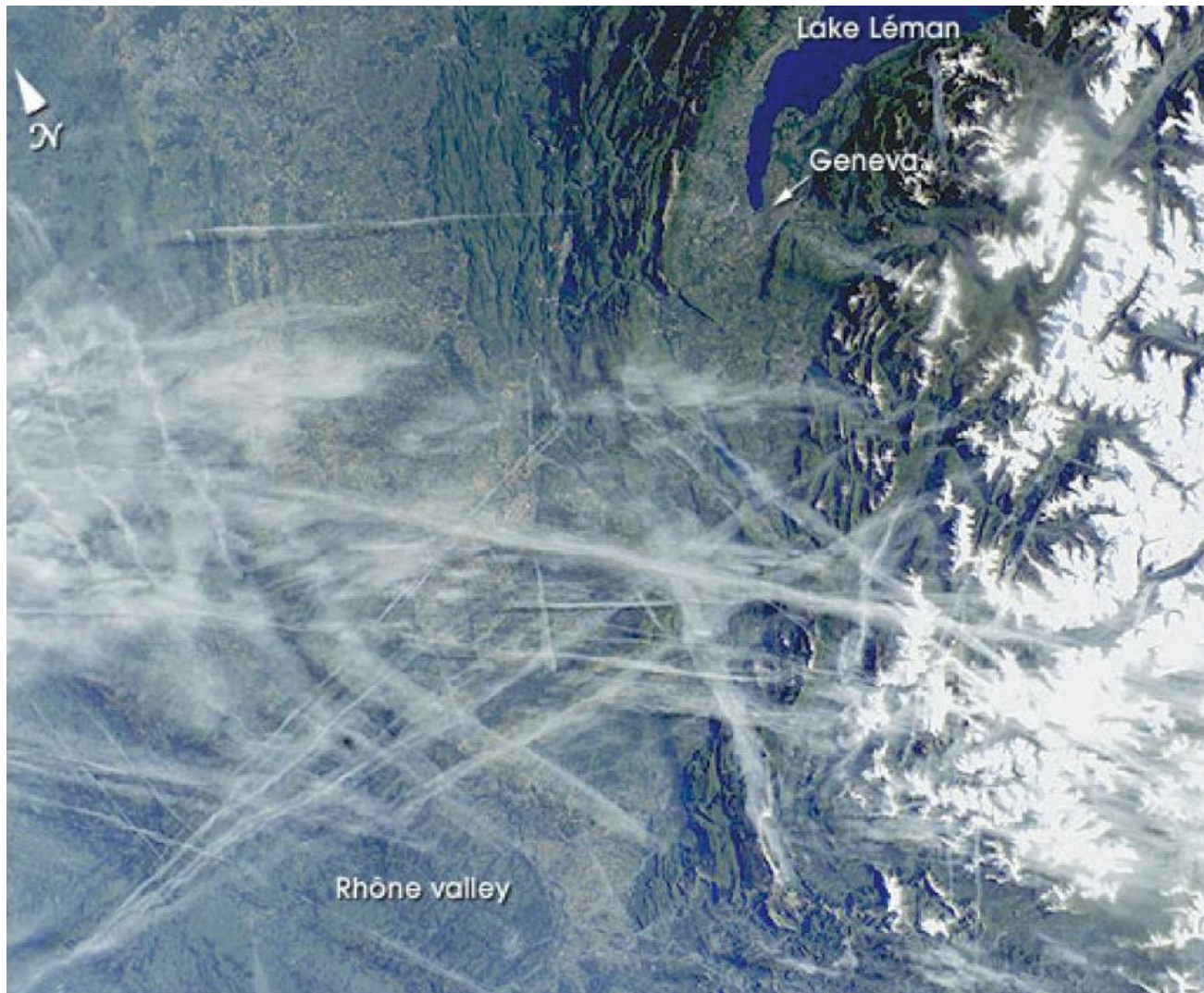


NASA, 2002
Atlantic, France, Spain



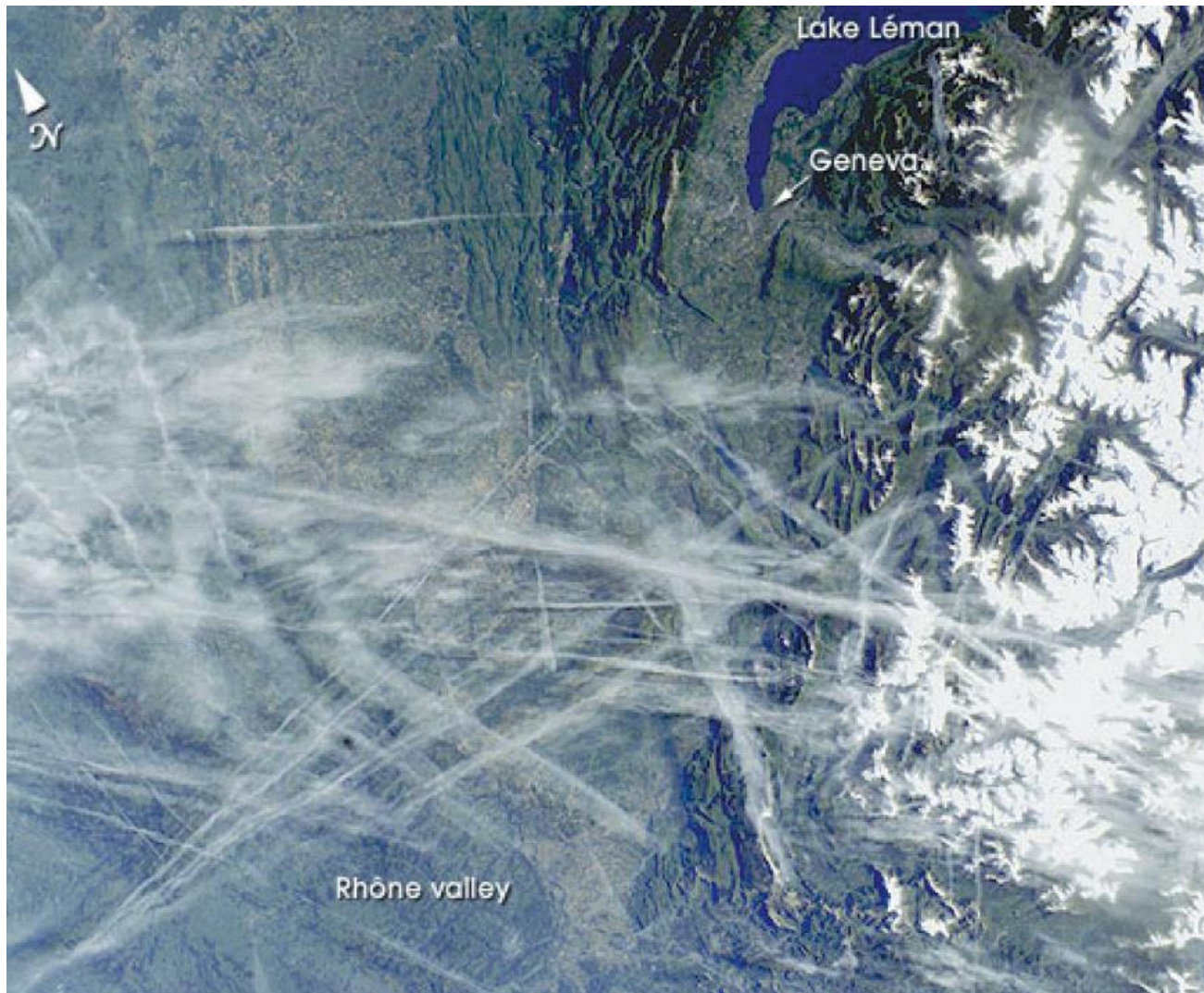
AVHRR, 27. Sept. 1987, 22:45 GMT
US-west coast

OTHER EVIDENCE OF CLOUD FORCING: CONTRAILS AND “AIRCRAFT CIRRUS”

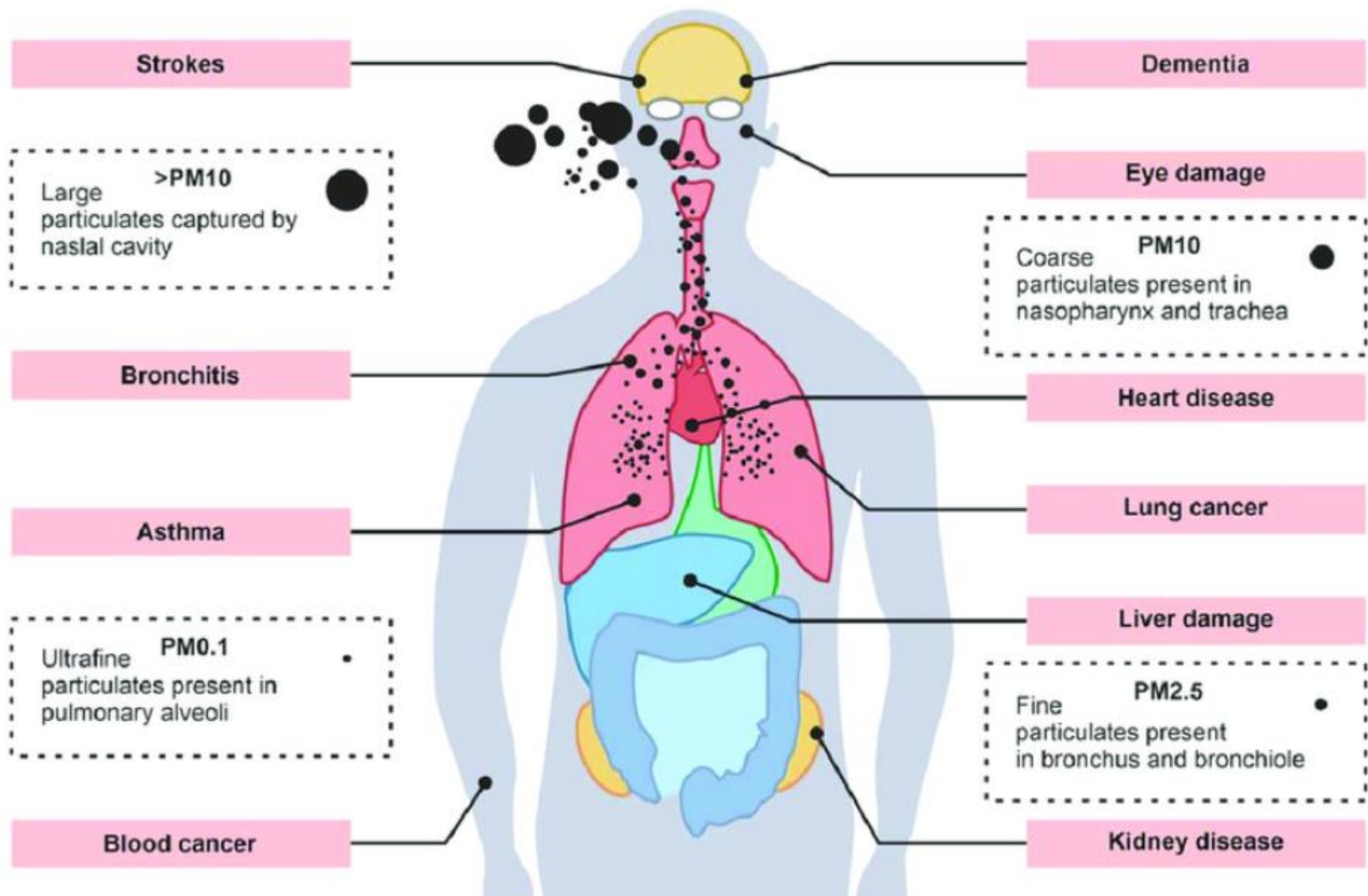


Aircraft condensation trails (contrails) over France, photographed from the Space Shuttle (©NASA).

OTHER EVIDENCE OF CLOUD FORCING: CONTRAILS AND “AIRCRAFT CIRRUS”



Aircraft condensation trails (contrails) over France, photographed from the Space Shuttle (©NASA).



AIR POLLUTION HUMAN HEALTH IMPACTS

EYES

Because there is a high flow of blood in the eyes, they are especially sensitive to small pollution particles like those found in PM2.5. Conditions such as dry eye syndrome, retinopathy, glaucoma, and cataracts have been connected to high air pollution exposure.



BRAIN

Air pollution exposure has been linked to a variety of neurological and cognitive impacts, including memory impairment, learning disabilities, anxiety, depression, schizophrenia, ADHD, and neurological conditions including dementia, Alzheimer's disease, Parkinson's disease, and stroke. Studies have even linked precise air pollution decreases to lowered dementia risk.



LUNGS

A slew of respiratory impacts are attributed to dirty air, from respiratory inflammation to asthma development to chronic loss of pulmonary function. Because most air pollutants are breathed in, the respiratory system is often the place where air pollution-related disease is most readily observed.



HEART

Cardiovascular disease and death are closely linked to air pollution, with outcomes of heart disease, heart failure, cardiac arrest, and arrhythmias. Some studies have even shown a stronger correlation between cardiovascular damage and death after air pollution exposure than observed with respiratory diseases.



STOMACH

Studies have demonstrated a link between poor air quality and gastrointestinal diseases, including inflammatory bowel disease (IBD), irritable bowel syndrome (IBS), and appendicitis, as the inhalation of air pollution is associated with changes to the gut microbiota. Long-term exposure to high concentrations of NO2 and PM has been linked to the early onset of Crohn's Disease.



LIVER

Animal studies find that long-term exposure to ambient air pollution is associated with metabolic-associated fatty liver disease, a disease that affects a quarter of the global population. The disease can progress to liver cancer and liver-related death.



BONES

A 2020 study found that ambient air pollution exposure is linked to lower bone mass. Other studies have found that air pollution may act as a risk factor for osteoporosis and be linked to higher rates of hospitalization for bone fractures, though research is limited.



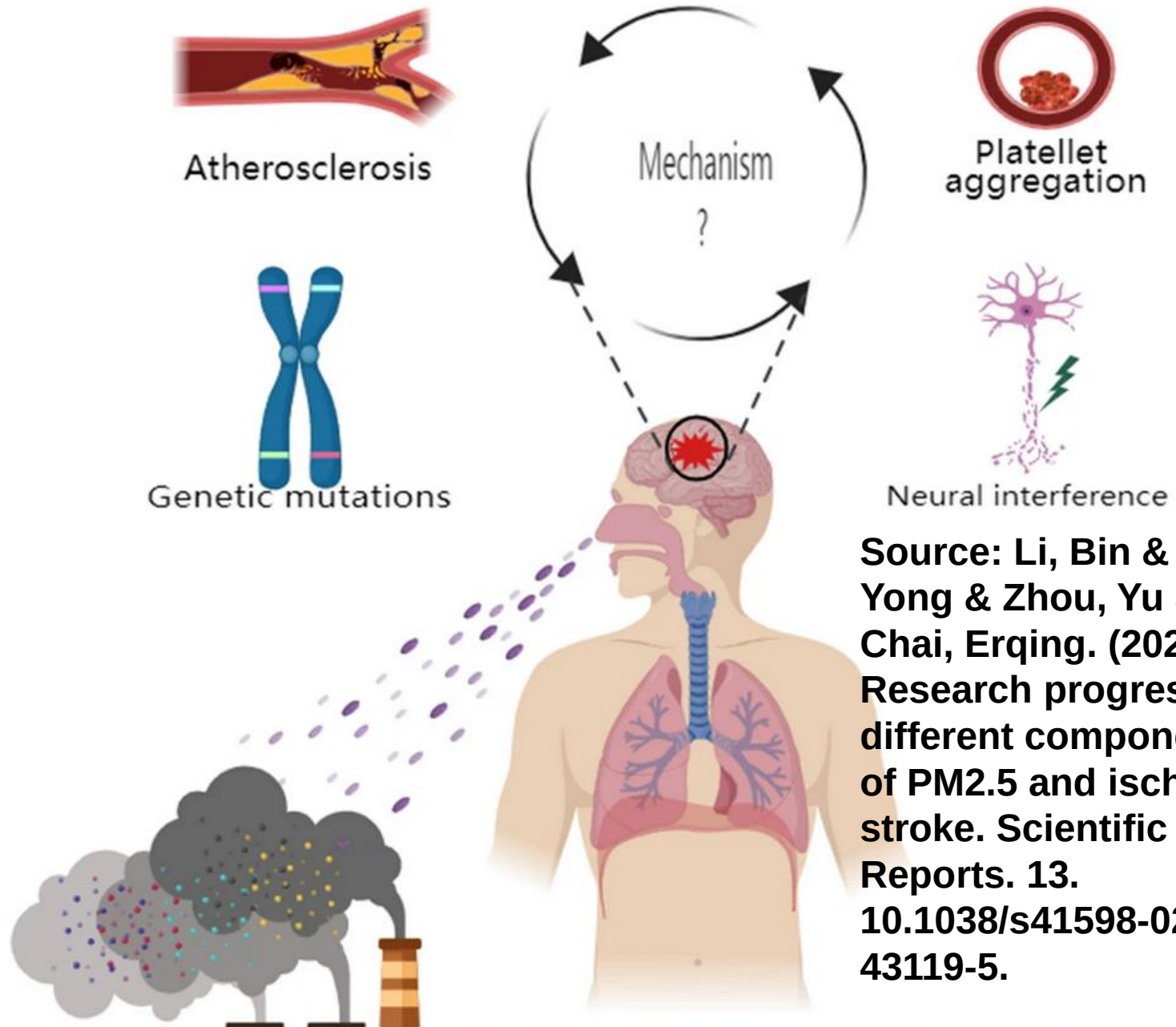
REPRODUCTION

Though the mechanisms behind it are not yet understood, exposure to higher levels of air pollution has been associated with lower levels of fertility and more difficulty in conceiving, including in those undergoing in vitro fertilization, as well as in a variety of sperm quality parameters. Further research is needed to investigate how exactly air pollution acts on the reproductive system.



Source: <https://www.istockphoto.com/vector/vector-medical-and-healthcare-infographic-gm804955246-130550021>

From: [Research progress of different components of PM_{2.5} and ischemic stroke](#)



Source: Li, Bin & Ma, Yong & Zhou, Yu & Chai, Erqing. (2023). Research progress of different components of PM_{2.5} and ischemic stroke. Scientific Reports. 13. 10.1038/s41598-023-43119-5.

Stokes Law

$$v = \frac{g d^2 (p_1 - p_2)}{18\eta}$$

Where:

$g = 9.80 \text{ m/s}^2$ the acceleration caused by gravity

d = diameter of particle in meters

p_1 = density of the particle expressed in g/m^3

p_2 = density of air in g/m^3

Because $p_1 \gg p_2$, then $(p_1 - p_2)$ is approximately equal to p_1 ,
and $\eta = 1.9 \times 10^{-2} \text{ g/ms}$, which is the viscosity of air at
atmospheric pressure and 25°C

Calculate the settling velocity of PM2 and also estimate its atmospheric residence time against deposition for PM2 released from 100 m stack height.

Solution:

$$v = \frac{gd^2\rho_p}{18\eta} = \frac{(9.80 \text{ m/s}^2) (2 \times 10^{-6} \text{ m})^2 (1.0 \times 10^6 \text{ g/m}^3)}{18(1.9 \times 10^{-2} \text{ g/ms})}$$

$$v = 114.6 \times 10^{-6} \text{ m/s}$$

$$v = 0.011 \text{ cm/s}$$

This particle has a settling velocity of 0.011 cm/s.

Part 2: The settling rate is the distance it will drop in a specific unit of time. The settling rate per day will be:

$$(8.6 \times 10^3 \text{ sec/day})(114.6 \times 10^{-6} \text{ m/s}) = 9.8 \text{ m/day}$$

If the fly ash particle described above was released into the atmosphere from a 100 meter tall smokestack it would take 10 days for it to settle on the ground. Stoke's law predicts that the residence time of a particle in air increases as the radius decreases. The residence times of particles in tropospheric air are often estimated by assuming they fall 10 m/day.